

1(a)	A – label line pointing to starch grain ; B – label line pointing to stroma ;	2
1(b)(i)	granum / grana ; A stack of thylakoids I thylakoid	1
1(b)(ii)	any four from: 1 (stack of) <u>thylakoids</u> ; 2 (membranes / thylakoids / C) form large / increase, surface area ; 3 for, (named) pigments / photosystems / light-harvesting clusters ; 4 for absorption of light <u>energy</u> ; 5 so, large number of / many, enzymes / ETC / ATP synthase / stalked particles ; 6 for, light dependent stage / photophosphorylation ;	4
1(c)(i)	1 oxygen, combines / reacts, with, rubisco / RuBP ; 2 less / no, carbon dioxide, combines / reacts, with, rubisco / RuBP ; A less or no carbon fixation 3 <i>ref. to photorespiration</i> ; <i>ignore refs to denaturation</i>	2

Question	Answer	Marks
1(c)(ii)	<i>any three from:</i> 1 photosynthesis (by C3 plants) is occurring ; 2 decrease in (atmospheric) carbon dioxide or increase in (atmospheric) oxygen or increase in / high, oxygen to carbon dioxide ratio ; 3 rubisco favours reaction with oxygen / AW ; 4 C4 plants have a selective advantage / description ; 5 oxygen acts as a selection pressure ; 6 AVP ; e.g. <i>ref. to</i> mutation in C3 plants	3